
Coarsening Confers Robustness: Causal Bayesian Optimization under DAG Misspecification

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Abstract

Causal Bayesian optimization (CBO) uses a causal graph to prune the space of candidate interventions and to build informative priors, but requires the entire DAG to be specified correctly, and robustness to structural misspecification is a recognized open problem of the field. We propose coarsening as an input contract: the practitioner supplies only a partition of the manipulable variables and the cluster-level (quotient) graph, and we prove an input-equivalence guarantee — any optimizer whose graph-dependent components factor through this quotient behaves identically under every fine-graph hypothesis with the same quotient, so no misspecification confined within a cluster can affect its exploration sets, priors, surrogates, or trajectory. We instantiate the contract for the three main families of the field: QCBO, which applies CBO’s own minimal-intervention-set rule to the C-DAG, identifiability deciding each arm’s prior tier (a do-calculus prior, or a common uninformative one) rather than its membership; QDCBO, which quotients dynamic CBO slice by slice; and QMCBO, which coarsens model-based CBO’s mechanism factorization into joint cluster mechanisms. Each is invariant, trajectory for trajectory, under within-cluster edge errors that measurably shift its full-graph counterpart, and each reduces exactly to its base method at the identity partition. The guarantee protects behavior, not solution quality: when the optimum splits a cluster, a fixed partition pays regret linear in the horizon, and plateau-triggered refinement down a declared hier-

archy restores convergence to the optimum. The trade is intrinsic: sub-cluster effects are provably not identifiable from quotient-level inputs alone — resolution must be bought with exposure.

1 Introduction

Causal Bayesian optimization [Aglietti et al., 2020] unifies two ideas: when the variable Y we wish to optimize sits inside a causal system, the right search space is the space of interventions $\text{do}(\mathbf{X}_s = \mathbf{x}_s)$, and observational data plus the causal graph yield informative priors over the interventional response surfaces. The catch is the input requirement: CBO and its descendants [Aglietti et al., 2021, 2023, Sussex et al., 2023] assume the full DAG is specified correctly, and the survey of Anonymous [2026] ranks robustness to causal-assumption violations first among the field’s open problems, with stress tests showing that method rankings can change substantially when only the assumed graph is perturbed. We study *graph misspecification*: the learner is given, and trusts, a single graph that differs from the truth (as opposed to graph uncertainty, where the learner maintains a posterior). Our thesis is that this fragility is best addressed by committing to less structure of a more reliable kind.

A wrong edge is costly: a two-variable example. The assumed graph enters a CBO algorithm only through its derived causal-effect estimators, and it corrupts them in two ways: it biases the do-calculus prior, and it can change the search set itself, possibly removing the candidate that contains the optimum (Remark 2). Either way the damage is a functional of the wrong edge: an estimator that cannot see an edge cannot be damaged by it. Fig. 1 makes this concrete on the smallest SCM in the paper: manipulable X_1, X_2 , latently confounded ($X_1 \leftrightarrow X_2$), both parents of $Y = \lambda(X_1 - 2)^2 + (X_2 + 2)^2 + \varepsilon$, so the uniquely optimal intervention is the joint arm $\text{do}(X_1=2, X_2=-2)$ with $y^* = 0$. Delete $X_1 \rightarrow Y$ in the assumed graph

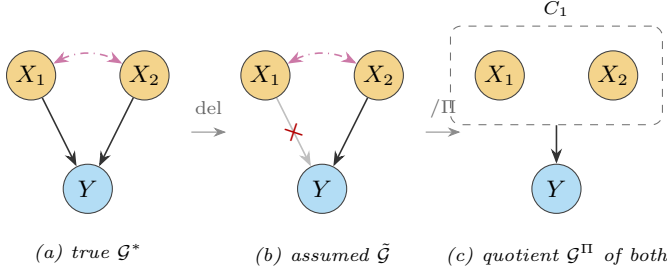


Figure 1: The paper in one figure (PARALLEL PARENT; orange: manipulable, blue: target, dash-dotted: latent confounding). Deleting $X_1 \rightarrow Y$ (b) removes every X_1 -containing arm from the fine exploration set — including the uniquely optimal joint arm — but is *invisible* in the quotient (c): $C_1 \rightarrow Y$ survives through $X_2 \rightarrow Y$. A full-DAG method permanently pays the calculable gap $G(\lambda) = 4.25\lambda$; a quotient method’s trajectory is identical under (a) and (b).

and X_1 leaves $\text{An}(Y)$: MIS pruning (Eq. 2) drops *every* X_1 -containing arm, and no amount of interventional data restores an arm the search set does not contain — full-DAG CBO plateaus permanently at the analytic gap $G(\lambda) = \lambda(4 + \text{Var}X_1) = 4.25\lambda$ (§5.1). Commit instead only to the cluster $C_1 = \{X_1, X_2\}$ and its quotient $C_1 \rightarrow Y$: the deletion is quotient-invisible ($C_1 \rightarrow Y$ survives through $X_2 \rightarrow Y$), so a quotient method’s exploration set, priors, and entire trajectory are *identical* under both hypotheses, and its lossless cluster arm still reaches y^* . The trade is equally immediate: edits *visible* in the quotient (say, an assumed $X_1 \leftrightarrow Y$) are not protected, and whole-cluster actions can be too coarse — when every optimum sets a strict subset of a cluster, a fixed partition pays a positive price (Prop. 3) that only refinement buys back (§5.3).

Coarser commitments decide less, so less can be wrong. A cluster-level structural belief encodes strictly fewer decisions than a full DAG: one aggregate judgement between two groups of variables stands in for a bundle of edge-level judgements the modeler is never asked to make. Both ends of this spectrum are served: full-DAG CBO at one end, methods that learn structure online at heavy per-trial cost at the other [Mukherjee et al., 2024, Branchini et al., 2023]. The practical middle, reliable cluster-level structure, has a developed identification theory [Anand et al., 2023, Ferreira and Assaad, 2025, Wahl et al., 2024] but, to our knowledge, no optimization method.

Quotient CBO. QCBO fills this gap. The practitioner supplies a partition Π of the manipulable variables and the C-DAG $\mathcal{G}^\Pi$ over its clusters (when a

fine-grained hypothesis graph is available, $\mathcal{G}^\Pi$ is its quotient, with bidirected edges absorbing latent confounding; Lee and Bareinboim, 2019). QCBO explores cluster-level interventions — the minimal intervention sets of the C-DAG, the same rule CBO applies to a full DAG — using identified cluster-level do-calculus estimates as GP prior means where the quotient identifies them, and a common uninformative prior where it does not. This input never looks inside a cluster: an edge error confined to a cluster cannot even be represented, let alone affect the optimizer.

Contributions.

- **Robustness as an input contract.** Quotient invariance is *method-agnostic* (Theorem 1): any optimizer whose graph-dependent components factor through $(\Pi, \mathcal{G}^\Pi)$ is invariant to arbitrary within-cluster edge errors. We instantiate the contract for the arm-based, dynamic, and model-based families (QCBO, QDCBO, QMCBO; §4).
- **Exploration focus.** Coarsening concentrates exploration on fewer cluster-level arms; Prop. 3 characterizes its cost.
- **Mechanism-isolating experiments.** Three deliberately minimal SCMs with closed-form oracles isolate the failure channels one at a time — exploration-set corruption (a calculable permanent gap), prior corruption (transient, and only where priors matter), and the price of coarsening with its recovery by refinement — followed by a portability suite running each quotient method against its base on the base paper’s own experiments (§5).

2 Background

A structural causal model $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, F, P(\mathbf{U}) \rangle$ induces a distribution over observables $\mathbf{V}$ and, for any $\mathbf{X}_s \subseteq \mathbf{V}$, interventional distributions $P(\cdot \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s))$ obtained by replacing the mechanisms of $\mathbf{X}_s$ with constants [Pearl, 2009]. The induced graph $\mathcal{G}$ has a directed edge per functional argument and a bidirected edge per shared latent; projecting out latents yields an ADMG over $\mathbf{V}$ [Lee and Bareinboim, 2019].

Causal Bayesian optimization. Given manipulable variables $\mathbf{X} \subseteq \mathbf{V}$ and target Y , CBO [Aglietti et al., 2020] solves

$$(\mathbf{X}_s^*, \mathbf{x}_s^*) = \arg \min_{\substack{\mathbf{X}_s \in \mathbf{ES} \\ \mathbf{x}_s \in D(\mathbf{X}_s)}} \mathbb{E}[Y \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s)], \quad (1)$$

where $D(\mathbf{X}_s)$ is the interventional domain, maintaining one GP per candidate set $\mathbf{X}_s$ in an exploration set $\mathbf{ES}$ (each a continuous BO sub-problem over $\mathbf{x}_s$), with

prior mean the do-calculus estimate from observational data and the assumed graph. The exploration set is pruned structurally: CBO searches over the *minimal intervention sets* (MIS) of the assumed graph [Lee and Bareinboim, 2018, Aglietti et al., 2020]. Writing $\mathcal{G}_{\mathbf{X}_s}$ for the mutilated graph with all edges into $\mathbf{X}_s$ removed and $\text{An}(\cdot)$ for ancestors,

$$\text{ES} = \text{MIS}(\mathcal{G}) = \{ \emptyset \neq \mathbf{X}_s \subseteq \mathbf{X} : \mathbf{X}_s \subseteq \text{An}(Y) \text{ in } \mathcal{G}_{\mathbf{X}_s} \} : \quad (2)$$

if some member of $\mathbf{X}_s$ is no ancestor of Y once the whole set is intervened on, $\text{do}(\mathbf{X}_s)$ induces the same distribution as intervening on a strict subset, so the arm is redundant. The possibly-optimal sets POMIS are a further graph-only pruning with $\text{POMIS} \subseteq \text{MIS}$ [Lee and Bareinboim, 2018]. Each trial selects the arm and value maximizing cost-normalized causal expected improvement [EI; Garnett, 2023]; the model-based family selects instead by GP-UCB with MCBO’s default exploration weight $\beta = 10$ [Sussex et al., 2023]. Both the pruning and the priors are functions of the assumed graph, which is where misspecification enters.

Two failure channels; the two-tier rule. A wrong assumed graph reaches the optimizer through exactly two channels (Remark 2): it *biases or forfeits priors* — the do-calculus prior mean of any arm whose estimand references the edge — and it can *change the exploration set itself*, removing an arm (possibly the optimal one) or admitting a spurious one. The two behave very differently: a corrupted prior is overridden by interventional data as long as the arm survives, while a removed arm is never played again. QCBO keeps the two channels separated by design: identifiability from the (quotient) graph decides an arm’s *prior tier* — an identified do-calculus prior, or a common uninformative one (the observational mean of Y) — but never its membership (Eq. 4, §4.1).

Coarsening and C-DAGs.

Definition 1 (Coarsening [Madaleno et al., 2026a]). *Given a DAG $\mathcal{G}$ on $\mathbf{V}$, a partition $\Pi = \{C_1, \dots, C_K\}$ of $\mathbf{V}$ induces the surjection $\chi : \mathbf{V} \rightarrow \Pi$ mapping each vertex to its cluster, and the quotient graph $\mathcal{G}^\Pi$ with vertex set Π and edge set $\{\chi(V_i) \rightarrow \chi(V_j) \mid V_i \rightarrow V_j \in \mathcal{G}, \chi(V_i) \neq \chi(V_j)\}$. If $\mathcal{G}^\Pi$ is acyclic, it is a **coarsening** of $\mathcal{G}$ and Π is valid.*

For ADMGs the quotient additionally carries $C_k \leftrightarrow C_\ell$ iff some $V_i \in C_k, V_j \in C_\ell$ share a bidirected edge. The result is a C-DAG in the sense of Anand et al. [2023]: do-calculus on it is sound and complete for the class of fine graphs that quotient to it. Markov properties transfer to the cluster level, faithfulness does not [Wahl et al., 2024].

Evaluation metrics. All metrics are direct functions of the trajectory and the oracle optimum y^* : the best value found y_T^{best} (final Y), the simple regret of the incumbent $r_t = |y_t^{\text{best}} - y^*|$, and the cumulative regret $R_T = \sum_{t=1}^T r_t$. Behavioral robustness is scored by *paired-by-seed* differences between a misspecified and a correct run of the same method (§5); solution quality by final Y and regret against the closed-form y^* .

3 What coarsening buys

Coarsening is not a property of one algorithm, so we state it as an input contract. An optimizer accepts the pair $(\Pi, \mathcal{G}^\Pi)$ in place of a fine graph, and every graph-dependent component (search space, identification formulas, priors, surrogate architecture, selection rules) is required to be a function of that pair. Any optimizer meeting the contract inherits the invariance below, whatever its family. Read the theorem for what it is: an *input-equivalence* statement. Two hypotheses with the same quotient are indistinguishable to the optimizer, so no intra-cluster mistake can change its behavior; the theorem does not certify solution quality, which rests on the quotient itself being correct. A wrong quotient produces the same trajectory for the wrong problem.

Throughout, Π, Π' are manipulable-only partitions. $\Pi_\perp$ is the finest (all-singleton) partition, the bottom of the partition refinement lattice [Madaleno et al., 2026a], and $\Pi' \preceq \Pi$ (Π' refines Π) iff every cluster of Π' is contained in a cluster of Π . Props. 1–3 relate quotients of a common underlying graph. The instantiations of §4 consume only $\mathcal{G}^\Pi$ (Assumption 1). The section is organized around three questions: what is protected (§3.1), what protection costs (§3.2), and why the two cannot be had for free (§3.3).

3.1 What is protected?

Theorem 1 (Quotient invariance is method-agnostic). *Let $\mathcal{A}$ be any causal Bayesian optimizer whose graph-dependent components (search space, identification formulas, priors, surrogate architecture, and selection rules) depend on the structural hypothesis $\mathcal{G}$ only through the pair $(\Pi, \mathcal{G}^\Pi)$, where $\mathcal{G}^\Pi$ is the quotient of $\mathcal{G}$ ’s latent projection under a valid Π . Then for any two acyclic hypotheses $\mathcal{G}, \mathcal{G}'$ with $(\mathcal{G}')^\Pi = \mathcal{G}^\Pi$, and the same data, seeds, and hyperparameters, $\mathcal{A}$ produces identical trajectories under $\mathcal{G}$ and $\mathcal{G}'$. In particular, this holds for every edit (insertion, deletion, or reversal of directed or bidirected edges) confined within a single cluster, and for inter-cluster edits that are redundant in the quotient. (Proof: App. A)*

Remark 1 (Exact scope). *A quotient-level input can-*

not represent an intra-cluster hypothesis, so no intra-cluster mistake can affect the optimizer. The precise condition is that the perturbed edge be invisible in the quotient: intra-cluster edges, and inter-cluster edges made redundant by a parallel edge. A quotient-visible edge is not protected (§5.1).

Remark 2 (How a wrong graph enters the algorithm). A wrong assumed graph reaches any causal Bayesian optimizer through its derived causal-effect estimators, in two ways. Prior bias: the do-calculus prior mean of any arm whose estimand references the edge is biased even though the true effect is unchanged, a convergence cost paid in regret; in QCBO a wrong graph can also flip an arm’s identifiability verdict and drop it to the uninformative prior tier (Eq. 4). Search-set change: the exploration set is itself derived from the graph, by MIS pruning (Eq. 2) in CBO [Lee and Bareinboim, 2018, Aglietti et al., 2020] and its C-DAG analogue (Eq. 3) in QCBO, so a wrong graph can remove an arm, possibly the optimal one, or admit a spurious one. Coarsening closes both channels whenever the edges are quotient-invisible (Theorem 1): the quotient never references the edge, so no estimator it forms is biased and no verdict it makes is flipped.

3.2 What does protection cost?

Proposition 1 (Validity is not refinement-monotone). *There exist a valid Π and a refinement $\Pi' \preceq \Pi$ whose quotient is cyclic; validity must be re-verified after each refinement (it remains decidable from cluster-level structure alone).*

Proposition 2 (Refinement: values are monotone, membership is not). *Let $\Pi' \preceq \Pi$ with both valid.* (i) *Every $\mathbf{X}_s \in \mathbf{ES}^\Pi$ that is again a MIS of $\mathcal{G}^{\Pi'}$ lies in $\mathbf{ES}^{\Pi'}$ with the same identification functional.* (ii) *Membership itself is not refinement-monotone: a coarse cluster-union arm need not be minimal at the finer partition. (On ToyGraph, $X \cup Z$ is the single MIS of the coarse quotient, while $\{X, Z\}$ is not a MIS at $\Pi_\perp$: mutilating Z cuts $X \rightarrow Z$, leaving X causally inert, so $\text{do}(X, Z) \equiv \text{do}(Z)$.) What refinement does preserve is value — Prop. 3 below — because every cluster-union arm’s value is attained by its MIS reduction (Lemma 1, App. A).*

Proposition 3 (Price of coarsening). *Let $\mu^*(\mathbf{X}_s) = \min_{\mathbf{x}_s \in D(\mathbf{X}_s)} \mathbb{E}[Y \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s)]$ and $V(\Pi) = \min_{\mathbf{X}_s \in \mathbf{ES}^\Pi} \mu^*(\mathbf{X}_s)$ — the best value reachable from the exploration set QCBO actually searches (Eq. 3). For any valid $\Pi' \preceq \Pi$, $V(\Pi_\perp) \leq V(\Pi') \leq V(\Pi)$, and $V(\Pi) = V(\Pi_\perp)$ iff some optimizer $\mathbf{X}_s^*$ of Eq. (1) is a union of Π -clusters. (Proof via Lemma 1: the MIS-restricted minimum equals the minimum over all cluster unions, which is refinement-monotone.)*

Proposition 4 (When coarsening is free). *Assume*

(i) *full-support intervention domains and (ii) cluster-contained confounding: in $\mathcal{G}^*$, every backdoor path from a cluster member to Y passes through another member of the same cluster. Then $V(\Pi) = V(\Pi_\perp)$ for every valid Π : setting the remaining members to the values they would naturally have taken changes nothing (consistency), so any sub-cluster arm’s value is a mixture over whole-cluster interventions, whose minimum the cluster arm attains. A union-of-clusters optimizer of Eq. (1) then always exists and the price of Prop. 3 is zero. Both premises are necessary (App. A: a policy-box example and a member–target confounding example, each with strictly positive price).*

Corollary 1 (The price is a finite-sample effect). *Under the premises of Prop. 4, if the exploration set contains a cluster arm covering an optimal fine arm and the arm-level optimizer is consistent with every arm sampled infinitely often, QCBO’s incumbent converges to y^* — whichever prior tier the arm runs on, since the uninformative tier only slows learning, never removes the arm. Coarsening then changes where finite samples concentrate, never the attainable value.*

Refining the partition: Hierarchical QCBO (HQCBO). When no optimizer of Eq. (1) is a union of Π -clusters, a fixed-partition run pays at least $V(\Pi) - y^* > 0$ per trial at every horizon: $R_T/T \geq V(\Pi) - y^*$ for all T — an action-space lower bound — with $R_T/T \rightarrow V(\Pi) - y^*$ under the additional condition that the base optimizer is consistent on the quotient’s arm set (Remark 4). Prop. 4 delimits when this can happen at all: a positive price requires a constrained intervention domain (the policy box of MEDIATEDCHAIN, §5.3) or confounding that escapes the cluster. Note the kind of failure this is. With a correct graph, a full-DAG method always retains an optimal arm, and a biased prior is eventually overridden by interventional data. A fixed coarse partition lacks the sub-cluster arm even when its quotient is correct: the gap is a property of the action space, and no amount of data restores it. (Under a wrong graph a full-DAG method can suffer the same uncorrectable arm loss through a flipped verdict, Remark 2; refinement addresses the action-space kind.) Refinement converts this failure back into a correctable one by treating Π as a starting resolution.

The practitioner may declare a *hierarchy* H : per cluster, an optional sub-partition (default: singletons); no new graph is supplied. HQCBO runs Algorithm 1 at Π and, when the incumbent improves by less than δ over k trials ($k=5$, $\delta=10^{-3}$, tuned once and fixed), splits the incumbent arm’s clusters via H , refusing any split that fails the validity re-check (Prop. 1). On acceptance the exploration set is rebuilt and unioned with every explored arm, data carry over keyed by arm

(Prop. 2), and only new arms receive the standard n_0 initialization points (App. C). We evaluate a single refinement step.

Remark 3 (Phase-scoped invariance). *Any misspecification that is quotient-invisible at every visited partition, in particular inside a cluster that is never split, leaves the entire HQCBO trajectory unchanged, including the trigger time and the refinement decision (per phase this is Theorem 1, and the trigger is a function of the trajectory and H alone). Refining a cluster spends its invariance: edits inside a split cluster are exposed after the trigger.*

Remark 4 (Refinement makes the price correctable). *The fixed-partition failure is worth stating mechanically: QCBO only ever executes whole-cluster interventions, so when every optimizer of Eq. (1) sets a strict subset of some cluster, every arm the fixed partition can play has value at least $V(\Pi) > y^*$. The per-trial regret is then bounded below by $V(\Pi) - y^* > 0$ at every t — no amount of data lowers it, because the deficit lives in the action space, not in the estimates — hence $R_T = \Omega(T)$ (and $\Theta(T)$, since per-trial regret is bounded on the compact intervention domains). Refinement changes the action space itself: once the exploration set at some visited partition contains an optimal arm and the base optimizer is consistent on a finite arm set — in particular, it samples every arm sufficiently often — the incumbent converges to y^* and the average regret $R_T/T \rightarrow 0$. We claim no convergence rate and report the post-trigger behavior empirically (§5.3). The trigger never observes y^* , so it also fires on lossless partitions; there refinement costs only the new arms’ initialization and, empirically, a small refinement price.*

Remark 5 (Splitting as coarse-to-fine MIS search). *MIS/POMIS pruning discards arms from the graph alone [Lee and Bareinboim, 2018]. HQCBO descends the partition lattice guided by observed value, stopping at the coarsest visited partition whose cluster-level MIS contains a value-optimal arm: a value-driven search for a partition exposing a minimal optimal intervention set, restricted to the declared hierarchy and the incumbent’s clusters.*

3.3 Why not both for free?

Could a quotient method keep its invariance *and* evaluate sub-cluster interventions when they matter? No: sub-cluster behavior is provably not a function of quotient-level inputs. Prop. 7 (§4.3) exhibits two SCMs with identical cluster mechanisms and identical whole-cluster interventional distributions that disagree on a sub-cluster intervention — so *no* method whose inputs factor through $(\Pi, \mathcal{G}^\Pi)$, arm-based, dynamic, or model-based, can distinguish them, and any method

Table 1: One recipe, three families: each base method’s graph-dependent components and their quotient replacements.

Base	Graph dependence	Quotient replacement
CBO	MIS arms; do-calculus priors	cluster MIS of the C-DAG; two-tier C-DAG priors
DCBO	per-slice arms; temporal mechanisms	cluster MIS per slice; cluster mechanisms per slice
MCBO	per-node GP mechanism factorization	joint multi-output cluster mechanisms

that could would have to consume intra-cluster structure, forfeiting Theorem 1 for exactly that structure. Refinement is therefore not a loophole but a controlled purchase:

Remark 6 (Refinement spends invariance, exactly where it refines). *Let HQCBO split cluster C into Π' -clusters at some trigger. Post-trigger, its graph-dependent components factor through $(\Pi', \mathcal{G}^{\Pi'})$, so Theorem 1 still applies at Π' : every edit invisible in $\mathcal{G}^{\Pi'}$ — in particular anything confined to a never-split cluster — remains protected, while edits inside C are now visible and can affect the post-trigger phase. Each split re-exposes exactly the split cluster’s intra-cluster structure and nothing else. (Proof: App. A, a corollary of Theorem 1 applied per phase.)*

4 Instantiations: QCBO, QDCBO, and QMCBO

The contract is met by one recipe: replace each graph-coupled component of a base optimizer with its quotient counterpart (Table 1). We instantiate it for the three main families, in increasing order of graph coupling: CBO (the graph prunes arms and builds priors), DCBO (it additionally carries information across time slices), and MCBO (it is the surrogate’s architecture). Figure 2 shows the structural input each family consumes and its quotient; estimator mechanics live in App. C and E.

4.1 Arm-based: quotient CBO (QCBO)

QCBO uses only cluster-level structure: a partition Π of the manipulable variables $\mathbf{X}$ and a C-DAG $\mathcal{G}^\Pi$ over its clusters and the singleton nodes. Y and every non-manipulable observable stay singletons, and mixing manipulable and non-manipulable variables in one cluster is disallowed, since intervening on a cluster (jointly setting all its members, $\text{do}(C_k = \mathbf{c}_k)$) must be executable. When the practitioner instead holds a fine-grained hypothesis graph $\mathcal{G}$, $\mathcal{G}^\Pi$ is its quotient after projecting out hidden confounders [Lee and Bareinboim, 2019]. QCBO itself never requires the fine graph. The exploration set is CBO’s own rule

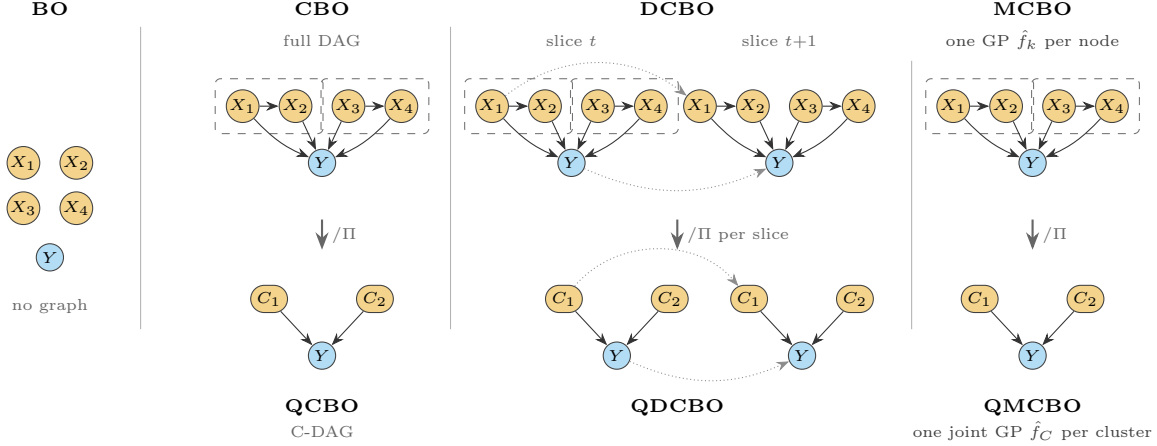


Figure 2: The input contract across the three families. Each base method (CBO, DCBO, MCBO) consumes a full graph; its quotient counterpart consumes only the partition Π and the C-DAG $\mathcal{G}^\Pi$ ($/\Pi$ arrows). Non-causal BO (left) consumes no graph. Intra-cluster edges — everything inside a dashed box — do not exist at the quotient level, which is the entire content of Theorem 1.

(Eq. 2) applied to the quotient: the MIS of $\mathcal{G}^\Pi$ over the manipulable cluster vertices Π_M ,

$$\mathbf{ES}^\Pi = \text{MIS}(\mathcal{G}^\Pi) = \left\{ \bigcup_{C_k \in \mathcal{S}} C_k \mid \emptyset \neq \mathcal{S} \subseteq \Pi_M, \mathcal{S} \subseteq \text{An}(Y) \right\} \quad (3)$$

with $\text{POMIS}(\mathcal{G}^\Pi) \subseteq \text{MIS}(\mathcal{G}^\Pi)$, so the possibly-optimal arms are always retained. Merging manipulable variables coarsens the action space: QCBO can only set a cluster jointly, never a strict subset of it. This is the mechanism behind both the robustness (§3) and the price (Prop. 3) of coarsening. Algorithm 1 (App. E) summarizes the loop.

Identifiability and the two-tier prior. For each arm $\mathbf{X}_s \in \mathbf{ES}^\Pi$ we pose the cluster-level query $P(Y \mid \text{do}(\mathbf{X}_s))$ on $\mathcal{G}^\Pi$ and run the complete ID algorithm of Shpitser and Pearl [2006] (via `ananke`; Ananke). Identifiability decides the arm’s *prior tier*, never its membership:

$$m_s(\mathbf{x}_s) = \begin{cases} \hat{\mathbb{E}}[Y \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s)] & \text{query id. from } \mathcal{G}^\Pi \\ & \text{and} \\ & \text{pipeline succeeds,} \\ \bar{Y}_{\text{obs}} & \text{otherwise.} \end{cases} \quad (4)$$

The identifiability test is exact where it applies: ID is complete on the C-DAG, and identification from $\mathcal{G}^\Pi$ is sound for every fine graph in its class [Anand et al., 2023]; a non-identifiable arm learns its response from experimental data alone. Estimator mechanics (the GP plug-in pipeline instantiating identified formulas) are in App. C.

Assumption 1 (Valid C-DAG; estimator boundary). *The directed part of $\mathcal{G}^\Pi$ is acyclic, which constrains*

only inter-cluster structure (intra-cluster acyclicity can even be relaxed; Madaleno et al., 2026b), coincides with validity of the coarsening when $\mathcal{G}^\Pi$ is a quotient, in $\mathcal{G}^\Pi$ is checkable on the cluster-level object. The ID test is complete, while the estimator pipeline behind Eq. (4) is sound but not complete: identifiable queries it misses receive the uninformative tier (in §5 this set is empty).

Proposition 5 (Identity recovery). *Under $\Pi = \Pi_\perp$, $\mathcal{G}^\Pi = \mathcal{G}^*$ and Eq. (3) is exactly Eq. (2): the exploration set is the MIS of the full projected graph and the priors are CBO’s do-calculus priors, so QCBO reduces to CBO. (On every benchmark graph of §5 each fine MIS arm is identifiable, so the uninformative tier is vacuous at $\Pi_\perp$. App. C documents the two places where the CBO authors’ shipped exploration sets deviate from the MIS definition — a non-minimal arm on ToyGraph and an ad-hoc size truncation on CoralGraph — and the corrected sets both methods use here.)*

4.2 Dynamic: quotient DCBO (QDCBO)

Dynamic CBO (DCBO; Aglietti et al., 2021) optimizes a sequence of interventions on a dynamic Bayesian network: each slice carries its own exploration sets and priors, built from within-slice (emission) and cross-slice (transition) mechanism estimates on the assumed temporal graph, and slice t ’s optimum feeds slice $t+1$ ’s priors. The graph enters the per-slice exploration sets, both estimate families, and the propagation structure.

Construction (QDCBO). Given the temporal graph over slices $t = 0, \dots, T-1$ and a partition Π of the per-slice manipulable variables, apply the recipe

per slice (Table 1): one vertex $C_{k,t}$ per cluster per slice (emission edges quotient within a slice, transition edges across; validity re-checked per slice, Prop. 1); exploration sets lifted to whole-cluster patterns (the slice-level analogue of Eq. 3); emission and transition estimates replaced by joint cluster mechanisms (the coregionalization construction of QMCBO below); and slice t 's optimum feeding slice $t+1$'s priors exactly as in DCBO, now at cluster level. Assumptions are DCBO's own (known temporal structure, causal sufficiency as used by its estimators) plus the contract's (manipulable-only clusters, per-slice validity). Singleton clusters reduce every component to stock DCBO.

Corollary 2 (QDCBO inherits the contract). *Every graph-dependent component of QDCBO is a function of $(\Pi, \mathcal{G}^\Pi)$ applied per slice, so Theorem 1 applies: quotient-invisible edits, in particular any edit confined to a cluster within a slice, leave its entire trajectory unchanged. At $\Pi = \Pi_\perp$, QDCBO is exactly DCBO, and its reachable intervention set at each slice equals QCBO's cluster-level arms, so the price of coarsening (Prop. 3) applies at each slice.*

4.3 Model-based: quotient MCBO (QMCBO)

Model-based CBO (MCBO; Sussex et al., 2023) couples the surrogate to the graph more tightly than CBO: it fits one GP per *mechanism*, evaluates interventions by composing these GPs along the DAG, and selects them by GP-UCB (§2). The graph is the surrogate's architecture, so a wrong intra-cluster edge corrupts the factorization every prediction flows through.

Construction (QMCBO). Coarsen the factorization: one *joint* mechanism per cluster, $P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})$, modeled as a multi-output GP over the cluster's member coordinates whose coregionalization (task) kernel couples them [Bonilla et al., 2007], composed along the C-DAG with joint within-cluster sampling. Intervention targets are lifted to whole-cluster patterns, and singleton clusters reduce to MCBO's per-node GPs.

Corollary 3 (QMCBO inherits the contract). *Every graph-dependent component of QMCBO is a function of $(\Pi, \mathcal{G}^\Pi)$, so Theorem 1 applies: quotient-invisible edits leave its entire trajectory unchanged. Moreover, (i) at $\Pi = \Pi_\perp$, QMCBO is exactly MCBO; and (ii) QMCBO has no arms, but its reachable intervention set (whole-cluster do patterns) equals QCBO's cluster-level arms, so Prop. 3 applies: on a lossy partition, no amount of data recovers y^* .*

Proposition 6 (Soundness of cluster mechanisms). *Assume causal sufficiency and a valid Π . For every*

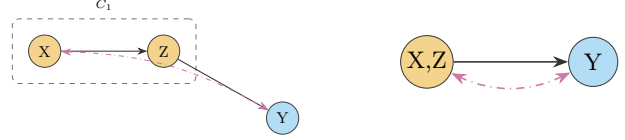


Figure 3: Coarsening can destroy identifiability the fine graph had. ToyGraph's fine DAG (left): $X \rightarrow Z \rightarrow Y$ with latent confounding $X \leftrightarrow Y$; $\text{do}(Z)$ is identifiable (backdoor through X) and $\text{do}(X)$ is identifiable (front-door through Z). Its quotient under $\Pi = \{\{X, Z\}\}$ (right) is a *bow*: $C_1 \rightarrow Y$ with $C_1 \leftrightarrow Y$, so $P(Y \mid \text{do}(C_1))$ is not identifiable from the C-DAG and the single arm runs on the uninformative prior tier (Eq. 4) — resolution bought with exposure (Prop. 7).

union-of-clusters intervention, the cluster-level truncated factorization (the true joint cluster mechanisms composed along the C-DAG with intervened clusters clamped) equals $P(\mathbf{V} \mid \text{do}(\mathbf{x}_S))$. (Proof: App. A.) The joint mechanism is essential: factoring a cluster into per-member single-output GPs discards within-cluster output correlation and is exact only when clusters contain no intra-cluster edges.

Proposition 7 (Exposure: sub-cluster interventions are not identifiable from cluster mechanisms). *There exist SCMs M_1, M_2 , causally sufficient and valid for the same Π , with identical joint cluster mechanisms and identical interventional distributions for every union-of-clusters intervention, yet $P^{M_1}(Y \mid \text{do}(\mathbf{x}_A)) \neq P^{M_2}(Y \mid \text{do}(\mathbf{x}_A))$ for a strict subset $A \subsetneq C$ of some cluster. (Proof: App. A.) Consequently no quotient-respecting method, arm-based, dynamic, or model-based, can evaluate sub-cluster arms from its inputs alone: the exposure trade of §3.2 is a property of the quotient contract itself, not of our refinement procedure. ToyGraph realizes the arm-based face of this trade (Fig. 3): merging $\{X, Z\}$ imports X 's latent confounding with Y into the single cluster arm — the C-DAG is the bow $C \rightarrow Y$, $C \leftrightarrow Y$ — so $P(Y \mid \text{do}(C))$ is not identifiable from the quotient even though $\text{do}(Z)$ is identifiable from the fine graph, and the arm runs on the uninformative prior tier (§5.4).*

5 Experiments

The experiments answer four questions, one mechanism at a time, on three deliberately minimal SCMs with closed-form oracles (MINIMALBENCH; specifications, per-condition exploration sets, and oracle derivations in App. B), then check that the contract carries across optimizer families on the field's own benchmarks. (i) Does quotient invisibility give exact invariance? (ii) Can that matter for optimization quality? (iii) What does a too-coarse partition cost?

(iv) Does the contract transfer to CBO, DCBO, and MCBO?

Protocol. Every condition holds the true SCM, the observational data ($n_{\text{obs}} = 100$ per run, one shared dataset per SCM), y^* , seeds, and trials fixed, and injects the misspecification only into the assumed structure the method reasons with, verifying that the objective is untouched (App. C). The methods share one estimator backend and differ only in the structure they consume: CBO identifies on the full assumed DAG over its MIS arms (by Prop. 5 it is QCBO at the finest partition), QCBO runs the same algorithm on the quotient (Eq. 3), non-identifiable arms on the uninformative tier. 30 seeds, $T=50$ trials (60 on MEDIATEDCHAIN), 3 initial interventional points per arm (a larger initial design on these 1–2D arms already contains near-optimal points, hiding the prior channel that §5.2 isolates), unit costs. Behavioral robustness is scored paired-by-seed: $\Delta_t = \text{metric}(\text{misspecified}, s, t) - \text{metric}(\text{correct}, s, t)$, so per-seed noise cancels; $\sup |\Delta|$ over trials and seeds is the invariance check. Table 2 classifies every perturbation in advance: whether the quotient changes, whether the fine MIS changes, whether fine prior tiers change, and hence whether protection is predicted — the experiments then confirm each row.

5.1 Experiment A: exploration-set corruption

PARALLEL PARENT (Fig. 1; $\lambda=1$) puts the optimum on the joint arm: fine MIS = $\{X_1\}, \{X_2\}, \{X_1, X_2\}$, $y^* = 0$ at $\text{do}(X_1=2, X_2=-2)$; the coarse partition $C_1 = \{X_1, X_2\}$ is lossless (its single cluster arm covers the optimum). Condition A1 deletes $X_1 \rightarrow Y$ from the assumed graph — quotient-redundant, since $C_1 \rightarrow Y$ survives through $X_2 \rightarrow Y$ — and the fine MIS collapses to $\{X_2\}$ alone: CBO loses the optimal arm *structurally*, before any data are seen. The damage is permanent and calculable: its best reachable value is $\min_{x_2} \mathbb{E}[Y \mid \text{do}(x_2)] = \lambda(4 + \text{Var}X_1) = 4.25$, and every seed’s full-horizon trajectory pins there (Fig. 4a) — the arm-deletion failure of Remark 2, realized at its analytic value. QCBO is trajectory-invariant: arm choices, intervention values, and incumbents are identical under A0 and A1 on every seed ($\sup |\Delta| = 0$, Table 2), and it reaches y^* . The negative control A3 assumes a spurious $X_1 \leftrightarrow Y$: this *is* quotient-visible (the C-DAG gains $C_1 \leftrightarrow Y$, a cluster bow), so QCBO keeps running — membership is never gated — but its cluster arm drops to the uninformative tier and its trajectory moves off its A0 path on every seed: the protection boundary is exactly where Remark 1 draws it. The intra-cluster bow A2 (add $X_1 \rightarrow X_2$ on the confounded pair) rounds out the taxonomy: it flips

$\text{do}(X_1)$ ’s fine prior tier, yet even *fine* CBO moves by $\sup |\Delta| < 10^{-2}$ here — the corrupted singleton is irrelevant to the optimum, which motivates Experiment B’s design.

5.2 Experiment B: prior corruption without arm loss

The recoverable channel needs the corrupted prior to sit on an arm that matters. FRONTDOOR arranges exactly that: $X_1 \rightarrow M \rightarrow Y$ with latent $X_1 \leftrightarrow Y$ and a *narrow-well* target $Y = 2(1 - e^{-(M-1)^2/2 \cdot 0.3^2}) + U + \varepsilon$, so the optimal arm is $\text{do}(M=1)$ ($y^* = 0$; backdoor through X_1), $\text{do}(X_1)$ is identified by the front-door functional, and the do-calculus prior’s head start — knowing where the well is — is the difference between sampling it immediately and searching a domain twenty well-widths wide. (On a smooth quadratic the same class of corruption measurably costs *nothing* — any BO self-corrects within a couple of trials, cf. the A2 discussion in §5.1. Prior corruption is a global-search phenomenon.) Condition B1 assumes a spurious intra-cluster confounder $X_1 \leftrightarrow M$: the fine MIS is untouched, but *both* fine arms — including the optimal $\text{do}(M)$ — lose identification and drop to the uninformative tier (Eq. 4). The effect is transient damage, precisely as Remark 2 classifies it: fine CBO under B1 is visibly slower early (Fig. 4b), paying $\Delta R_{50} = 6.1 \pm 1.0$ cumulative regret (paired, mean $\pm$ s.e., 30 seeds), yet recovers essentially the same final value (0.006 vs 0.001, against an observational baseline of 1.64) — the arm is still there, and interventional data override the lost prior. The coarse partition $\{X_1, M\}$ quotients to the ToyGraph bow ($C_1 \rightarrow Y$, $C_1 \leftrightarrow Y$), so QCBO’s single cluster arm runs on the uninformative tier *in both conditions* — and its trajectories, arm choices, and intervention values are exactly identical ($\sup |\Delta| = 0$): the assumed $X_1 \leftrightarrow M$ is intra-cluster, and the quotient cannot even represent it.

5.3 Experiment C: the price of coarsening, and buying it back

MEDIATEDCHAIN makes the price of Prop. 3 concrete in two variables: $X_1 \rightarrow X_2 \rightarrow Y$ with $Y = (X_2 - 4)^2$, $X_2 = 2X_1$ (plus noise), and the *policy box* $D(X_2) = [-2, 2]$. Fine MIS = $\{X_1\}, \{X_2\}$ — on a chain the joint is not minimal (Prop. 2) — and the fine optimum $\text{do}(X_1=2)$ drives X_2 ’s natural response to 4, *outside* the box: exactly the constrained-domain mechanism by which Prop. 4’s premise (i) fails. The whole-cluster arm $\text{do}(X_1, X_2)$ must clamp X_2 into the box, so $V(\Pi) = (2 - 4)^2 = 4$ against $y^* = \sigma_2^2 = 0.04$: an analytic price of 3.96. The runs realize it exactly (Fig. 5): CBO descends to y^* through its sub-cluster arm (final 0.040), fixed-partition QCBO pins at 4.000

Table 2: Perturbation taxonomy of the minimal suite, emitted from the same structural module the methods consume (so paper and code cannot drift). “Protected” is the *prediction* of Theorem 1 (quotient-invisibility); the last column is the measured paired invariance check $\sup_{s,t} |\Delta|$ for QCBO over 30 seeds — exactly 0 on every protected row, nonzero only on the quotient-visible negative control. PP = PARALLELPARENT, FD = FRONTDOOR.

Cond.	Edit (assumed graph)	Quotient changed	Fine MIS changed	Fine prior changed	Protected	$\sup \Delta $
A1 (PP)	del $X_1 \rightarrow Y$ (quotient-redundant)	×	✓	—	✓	0
A2 (PP)	add $X_1 \rightarrow X_2$ (intra-cluster bow)	×	×	✓	✓	0
A3 (PP)	assume $X_1 \leftrightarrow Y$ (quotient-visible)	✓	×	✓	×	21.4
B1 (FD)	assume $X_1 \leftrightarrow M$ (intra-cluster conf.)	×	×	✓	✓	0

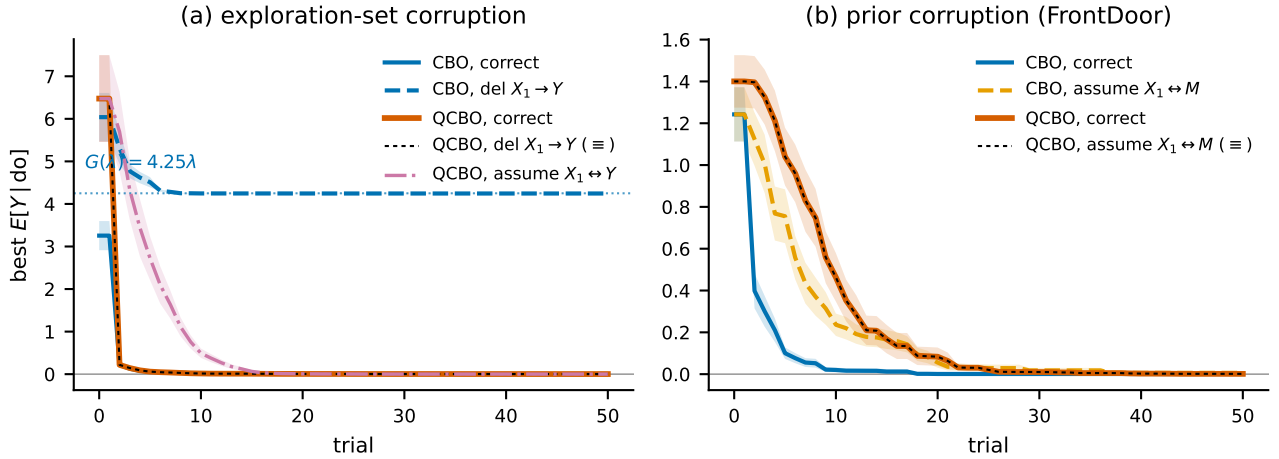


Figure 4: Correct (solid) vs misspecified (dashed) trajectories, mean $\pm$ s.e. over 30 seeds. (a) Exploration-set corruption (PARALLELPARENT): wrong-fine CBO plateaus at the analytic gap $G(\lambda) = 4.25$; QCBO’s misspecified trajectory coincides exactly with its correct one (black dashes over orange, $\sup |\Delta| = 0$); the quotient-visible control (pink) moves QCBO off its path. (b) Prior corruption (FRONTDOOR): the assumed intra-cluster $X_1 \leftrightarrow M$ costs fine CBO its do-calculus head start — transiently slower, same final value — while QCBO is exactly invariant.

on every seed and accrues cumulative regret at the linear rate the action-space bound prescribes (Remark 4; $R_{60} = 252.7 \pm 1.8$), and HQCBO — same coarse configuration, singleton hierarchy declared — plateaus, triggers at trial 7.5 ± 0.1 (30/30 seeds, no refused splits), splits $\{X_1, X_2\}$, and descends to 0.040 ± 0.000 : the fixed partition’s linear regret flattens once the refined exploration set contains the optimal fine arm ($R_{60} = 46.7 \pm 2.0$, against CBO’s 10.4 ± 3.0). Phase 1 is byte-identical to QCBO’s prefix on every seed (Remark 3).

5.4 Experiment D: portability across families

The minimal suite isolates mechanisms; the remaining question is whether the contract survives contact with the field’s own benchmarks and optimizer fami-

lies. We compare each quotient method with its base on the experiments the base method’s own paper ran, in that paper’s own metric, running every base through its authors’ released stack: CBO vs QCBO on the 2020 paper’s Toy, Synthetic, and coral-reef examples (best $E[Y | do]$, 10 seeds); DCBO vs QDCBO on its stationary, independent, and nonstationary DBNs at $T=3$ (best Y per slice, 20 seeds); MCBO vs QMCBO on its ToyGraph and PSAGraph tasks (best reward, 5 seeds; the paper’s remaining problems are function networks, where a partition of function nodes has no causal reading). Every dataset’s DAG and quotient are drawn in App. D; Fig. 6 overlays the trajectories and Table 3 reports finals and realized prices.

Value. The quotient matches or beats its base on six of the eight experiments, including QDCBO on all

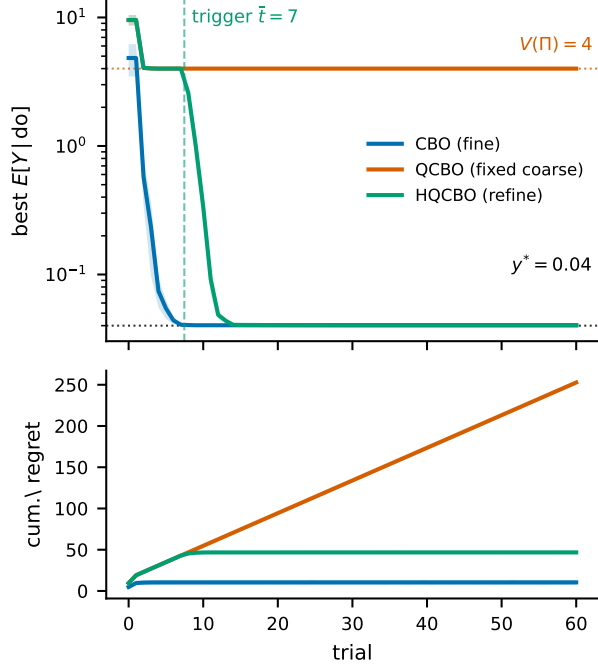


Figure 5: The price of coarsening and its recovery (MEDIATEDCHAIN, mean $\pm$ s.e., 30 seeds). Top: best-so-far against the analytic $V(\Pi) = 4$ and $y^* = 0.04$ (log scale); the vertical line marks the mean refinement trigger. Bottom: cumulative regret — linear growth for the fixed partition, flattening for HQCBO, with fine CBO the best constant.

three of DCBO’s own setups and QMCBO essentially reaching ToyGraph’s optimum (2.150 ± 0.013 vs 2.172 , exact on three of five seeds) where MCBO never improves on its initial design (1.14 ± 0.27) — the within-cluster correlation of Prop. 6 is the missing ingredient; on PSAGraph both converge to the same plateau (-5.15 ± 0.00), QMCBO in $3.0\times$ less wall-clock. The two material positive prices are exactly where Prop. 3 predicts them: Synthetic, whose optimum $\text{do}(B, D)$ splits the domain partition, and, mildly, Coral. On Toy the single coarse arm is the bow of Fig. 3: per the two-tier rule QCBO keeps the arm on the uninformative prior and still matches CBO (-2.13 ± 0.03 vs -2.16 ± 0.00).

Invariance within each family. Under within-family quotient-invisible perturbations of the model’s view, QMCBO’s trajectories are identical to their correct-graph runs on 10/10 (env, seed) pairs (Cor. 3 realized) while MCBO’s shift on 8/10 — in a direction that is not predictable: on these SCMs the wrong graph sometimes *helped*; QDCBO is likewise invariant on its three setups under the intra-slice edit while

Table 3: Realized price of coarsening per family, each pair on the base method’s own experiment suite (final Y , mean $\pm$ s.e.). Price is signed so that positive means the quotient did worse in the task’s direction.

Family	Experiment	Base final	Quotient final	Price	Task
CBO vs QCBO	Toy	-2.16 ± 0.00	-2.13 ± 0.03	$+0.03$	min
CBO vs QCBO	Synthetic	-3.14 ± 0.19	-1.25 ± 0.07	$+1.89$	min
CBO vs QCBO	Coral	36.08 ± 0.02	36.42 ± 0.01	$+0.34$	min
DCBO vs QDCBO	stat ($T=3$)	-6.14 ± 0.05	-6.43 ± 0.02	-0.29	min
DCBO vs QDCBO	ind ($T=3$)	-3.12 ± 0.06	-5.57 ± 0.15	-2.45	min
DCBO vs QDCBO	nonstat ($T=3$)	8.03 ± 0.90	6.33 ± 0.29	-1.69	min
MCBO vs QMCBO	ToyGraph	1.14 ± 0.27	2.15 ± 0.01	-1.01	max
MCBO vs QMCBO	PSAGraph	-5.15 ± 0.00	-5.15 ± 0.00	-0.00	max

DCBO moves (App. F). Which case a practitioner is in cannot be known without the very structure in doubt; the contract removes the question for every quotient-invisible error.

6 Related work

QCBO occupies an unfilled cell in the CBO landscape: it neither assumes the full DAG nor learns one, but takes a coarser, more reliable input and identifies on the quotient (Table 5, App. F). Methods relaxing the full-graph requirement learn structure online at per-trial cost (GACBO maintains a DAG posterior [Mukherjee et al., 2024], CEO hedges a graph posterior through an entropy-based acquisition [Branchini et al., 2023], Durand et al. [2025] identify $\text{Pa}(Y)$ under a Y -leaf assumption) or assume a known equivalence class [Park et al.]. Averaging only *dilutes* a wrong edge in proportion to posterior mass, whereas the quotient pays nothing per trial and is exactly invariant to any edit it does not resolve. The identification machinery we rely on is established (C-DAG do-calculus [Anand et al., 2023], C-DMGs over ADMGs [Ferreira and Assaad, 2025], cluster-level Markov properties [Wahl et al., 2024], relaxed admissibility [Bytez.com et al., 2025], C-DAGs as discovery background knowledge [Padh and Kilbertus]), but none of it performs optimization. Closest in spirit, Zeitler [2025] coarsen the *outcome* as a cheap proxy [Poloczek et al., 2017, Beckers and Halpern, 2019]. Causal abstraction relates decisions across levels of granularity, and the quotient is such a map, applied to the manipulable *inputs*.

7 Discussion and limitations

Theorem 1 is structural: coarsening acts as a hard regularizer on the hypothesis space, trading resolution (Prop. 3) for immunity to a class of misspecifications. Caveats. (i) The invariance belongs to the *quotient*, and the guarantee is about *behavior*: robustness is read off the CBO-to-QCBO contrast. (ii) DCBO and MCBO cannot join the fixed-objective seam (per-

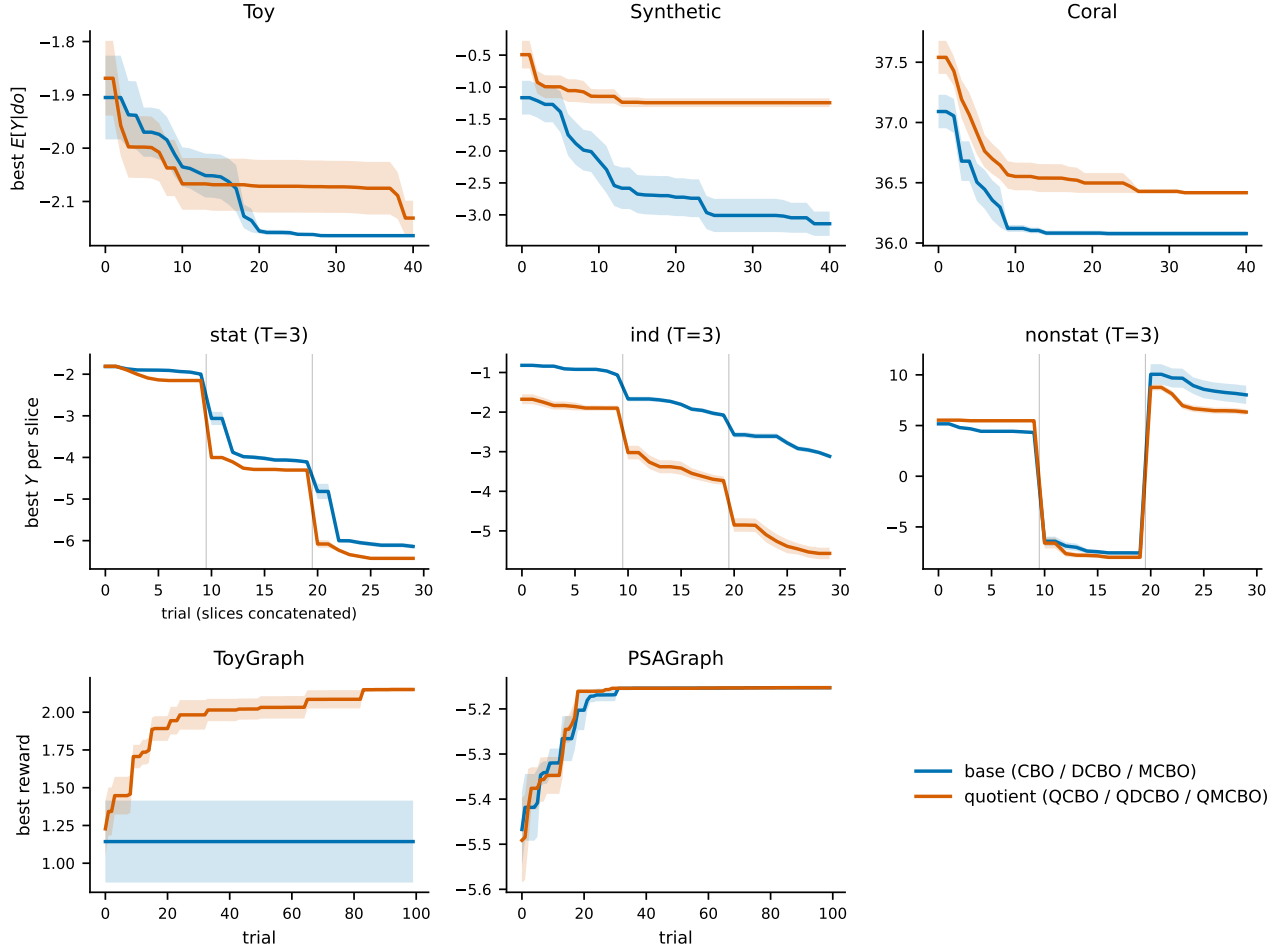


Figure 6: Each quotient method against its base on the base paper’s own experiments and metric (mean $\pm$ s.e.). Top: CBO vs QCBO (best $E[Y | do]$, minimization). Middle: DCBO vs QDCBO (best Y per time slice, trials concatenated across slices). Bottom: MCBO vs QMCBO (best reward, maximization).

environment coupling), so QDCBO and QMCBO are each evaluated within their own family’s harness with the base method’s analytic objective held fixed. (iii) Faithfulness does not transfer to the cluster level [Wahl et al., 2024]. (iv) Unit intervention costs only. (v) Building the exploration set costs one MIS test and one complete-ID check per candidate cluster union, up to $2^K - 1$ of each on the *quotient* — exponential in clusters, not in variables, and trivial at the cluster counts practitioners assert. (vi) QMCBO assumes causal sufficiency (Prop. 6), as MCBO does. (vii) What self-corrects and what does not: a wrong *prior* can eventually be overcome by interventional data provided the optimal arm remains in the exploration set; a wrong *exploration set* need not self-correct — under a wrong graph, full-DAG CBO’s MIS can permanently exclude the optimal arm (Remark 2), and under a lossy partition QCBO lacks the sub-cluster arm by construction

(Prop. 3). Neither failure is repaired by data; refinement repairs the second, and quotient-invisibility prevents the first for intra-cluster errors. A rollback rule bounding the lossless-trigger tax, and discovering the hierarchy from the run’s own data, are natural follow-ups.

8 Conclusion

Coarsening runs causal Bayesian optimization on a coarser, more reliable structural commitment with provable invariance to intra-cluster misspecification, at a price that is zero under the conditions of Prop. 4 and, otherwise, correctable by refinement — the fixed partition’s linear-in- T regret becomes $R_T/T \rightarrow 0$ once a visited partition’s exploration set contains an optimal arm and the base optimizer is consistent on it (Remark 4). The practitioner dials resolution against

robustness by choosing how much structure to commit to, in any of the three families.

Acknowledgements

A Proofs

Notation is as in §4.1 and §3.

Proof of Theorem 1. First, $(\mathcal{G}')^\Pi = \mathcal{G}^\Pi$: the latent projection removes only latent vertices, and an edge between two observables is never an interior vertex of a projection path, so $(\mathcal{G}')^*$ and $\mathcal{G}^*$ differ exactly by the modified intra-cluster edges. The quotient keeps a directed (resp. bidirected) cluster-level edge iff some fine edge crosses two distinct clusters, so edits confined within clusters, or redundant across them, leave it unchanged. It remains to show trajectory equality. Write $\mathcal{A}$'s run as a sequence of decisions $d_1, d_2, \dots$ (which arm/action to evaluate, model refits, stopping). By hypothesis each d_t is a deterministic function $d_t = f_t((\Pi, \mathcal{G}^\Pi), \mathcal{D}_{t-1}, \omega_t)$ of the quotient, the data collected so far, and the RNG stream. Under $\mathcal{G}$ and $\mathcal{G}'$ the first arguments coincide. The objective (the true SCM) and the seeds coincide by the fixed-objective protocol, so $\mathcal{D}_0, \omega_1$ coincide, hence d_1 coincides, hence $\mathcal{D}_1$ coincides. Induction over t gives equality of the entire trajectory. $\square$

Proof of Prop. 6. Under causal sufficiency the fine SCM is Markov relative to $\mathcal{G}$ with mutually independent exogenous noises, so for any intervention $\text{do}(\mathbf{x}_S)$ on a union of clusters S the fine truncated factorization gives $P(\mathbf{V} \mid \text{do}(\mathbf{x}_S)) = \prod_{k \notin S} P(X_k \mid \mathbf{X}_{\text{pa}_{\mathcal{G}}(k)})|_{\mathbf{x}_S = \mathbf{x}_S}$. Group the factors by the clusters of Π . For a cluster $C \not\subseteq S$, the product of its members' factors is, by the chain rule applied in any topological order of $\mathcal{G}$ restricted to C , exactly the joint conditional $P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})$, where $\text{pa}(C)$, the union of C 's parent clusters, contains every out-of-cluster variable appearing in a member's conditioning set (validity of Π guarantees the quotient is acyclic, so $\text{pa}(C)$ is well defined and precedes C in C-DAG order). The fine factorization therefore equals the product over clusters of the joint cluster mechanisms with the clusters of S clamped at $\mathbf{x}_S$: the cluster-level truncated factorization. Note both uses of sufficiency: independent noises across clusters (no bidirected edges) and the absence of unobserved common causes that would make a member's conditional depend on variables outside $\text{pa}(C) \cup C$. Replacing $P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})$ by the product of its per-member marginals would be exact iff the members are conditionally independent given $\text{pa}(C)$, in particular when C contains no intra-cluster edges — hence the joint mechanism in the construction. $\square$

Proof of Prop. 7. Take $\mathbf{X} = \{X_2, X_3\}$ as one root cluster C , target $Y = X_2 + \varepsilon_Y$ its child, all noises standard normal and independent (sufficiency holds, and any Π clustering $\{X_2, X_3\}$ is valid). Let M_1 : $X_2 = \varepsilon_2$, $X_3 = aX_2 + \varepsilon_3$; and M_2 : $X_3 = \varepsilon'_3$, $X_2 = a'X_3 + \varepsilon'_2$, with $a' = a/(1+a^2)$, $\text{Var}(\varepsilon'_3) = 1+a^2$, and $\text{Var}(\varepsilon'_2) = 1-a^2/(1+a^2)$. Both induce the same zero-mean Gaussian joint on (X_2, X_3) , with covariance $\begin{pmatrix} 1 & a \\ a & 1+a^2 \end{pmatrix}$, hence the same joint cluster mechanism (here, the same root-cluster distribution), and the same $P(Y \mid \text{do}(\mathbf{x}_C))$ for the whole-cluster intervention (both set X_2 directly). But under the sub-cluster intervention $\text{do}(X_3 = x)$: in M_1 , $X_2 = \varepsilon_2$ is unaffected, so $\mathbb{E}[Y \mid \text{do}(X_3=x)] = 0$ for all x ; in M_2 , $X_2 = a'x$, so $\mathbb{E}[Y \mid \text{do}(X_3=x)] = a'x \neq 0$ for $x \neq 0$, $a \neq 0$. Every quotient-respecting input (the C-DAG, the joint cluster mechanisms, all whole-cluster interventional distributions) coincides between M_1 and M_2 , yet the sub-cluster query differs. No function of those inputs can answer it. $\square$

Proof of Prop. 5. Singleton clusters make the quotient map the identity on $\mathcal{G}^*$, so $\mathcal{G}^\Pi = \mathcal{G}^*$, unions of singleton clusters are exactly the subsets of $\mathbf{X}$, and the MIS condition of Eq. (3) coincides verbatim with Eq. (2). The prior construction poses the same do-calculus queries CBO poses on the full graph, so priors coincide arm-for-arm. $\square$

Lemma 1 (MIS reduction). *Let G be an ADMG with manipulable vertices M and target Y , let $\emptyset \neq \mathbf{W} \subseteq M$, and set $\mathbf{W}' = \mathbf{W} \cap \text{An}(Y)$ computed in $G_{\mathbf{W}}$. Then (i) $P(Y \mid \text{do}(\mathbf{W} = \mathbf{w})) = P(Y \mid \text{do}(\mathbf{W}' = \mathbf{w}|_{\mathbf{W}'}))$ for every $\mathbf{w}$; (ii) $\mathbf{W}'$ is a MIS (or empty, when no member reaches Y); and (iii) consequently $\min_{\mathbf{x}_s \in \text{MIS}(G)} \mu^*(\mathbf{x}_s) = \min_{\emptyset \neq \mathbf{W} \subseteq M} \mu^*(\mathbf{W})$: restricting the search to MIS loses no value.*

Proof. (i) In $G_{\mathbf{W}}$ the members of $\mathbf{W} \setminus \mathbf{W}'$ have no directed path to Y , so Y is unaffected by their values under the intervention; dropping them from the intervention changes no mechanism on any directed path into Y (do-calculus rule 3). (ii) A directed path from $w' \in \mathbf{W}'$ to Y in $G_{\mathbf{W}}$ passes through no intervened vertex (their incoming edges are cut), hence through no member of $\mathbf{W} \supseteq \mathbf{W}'$; the same path exists in $G_{\mathbf{W}'}$, which only restores edges, so $\mathbf{W}' \subseteq \text{An}(Y)$ there. (iii) By (i) and (ii) every candidate's value is attained by a MIS, and $\text{MIS}(G)$ is a subset of the candidates. $\square$

Proof of Prop. 1. Take $\mathbf{X} = \{a_1, a_2, b_1, b_2\}$ with fine edges $a_1 \rightarrow b_1$ and $b_2 \rightarrow a_2$. The coarsest partition $\Pi = \{\{a_1, a_2, b_1, b_2\}\}$ is valid: both edges are intra-cluster, so the quotient has a single manipulable node and is acyclic. Its refinement $\Pi' = \{\{a_1, a_2\}, \{b_1, b_2\}\}$ exposes both edges as inter-cluster ($a_1 \rightarrow b_1$ contributes $A \rightarrow B$ and $b_2 \rightarrow a_2$ contributes $B \rightarrow A$), inducing the 2-cycle $A \rightleftarrows B$, so $\mathcal{G}^{\Pi'}$ is cyclic and Π' is invalid. Decidability is immediate from the acyclicity clause of Def. 1. $\square$

Proof of Prop. 2. (i) Membership at Π' is the stated hypothesis (any union of Π -clusters is a union of Π' -clusters, so the candidate exists at Π' ; being a MIS of $\mathcal{G}^{\Pi'}$ puts it in $\mathbf{ES}^{\Pi'}$). For the identification functional: quotienting is compositional — $\mathcal{G}^{\Pi}$ is the quotient of $\mathcal{G}^{\Pi'}$ under the induced partition of Π' -clusters — so every fine graph compatible with $\mathcal{G}^{\Pi'}$ is compatible with $\mathcal{G}^{\Pi}$. Identifiability from $\mathcal{G}^{\Pi}$ means a single functional of the observational distribution equals $\mathbb{E}[Y \mid \text{do}(\mathbf{X}_s)]$ across all SCMs whose graph is compatible with $\mathcal{G}^{\Pi}$ [Anand et al., 2023]. A fortiori the same functional works across the smaller class compatible with $\mathcal{G}^{\Pi'}$. (ii) The ToyGraph counterexample is stated in the proposition; nothing further is required. $\square$

Proof of Prop. 3. By Lemma 1(iii) applied to $\mathcal{G}^{\Pi}$, $V(\Pi) = \min_{\mathbf{X}_s \in \mathbf{ES}^{\Pi}} \mu^*(\mathbf{X}_s)$ equals the minimum of μ^* over all non-empty unions of Π -clusters (the value μ^* of a cluster union is a property of the underlying SCM, the same at every partition that can express the union). Unions of Π -clusters are unions of Π' -clusters and, at $\Pi_{\perp}$, all non-empty subsets of $\mathbf{X}$, so the three minima are over nested candidate families: $V(\Pi_{\perp}) \leq V(\Pi') \leq V(\Pi)$. Equality $V(\Pi) = V(\Pi_{\perp})$ holds iff the unrestricted minimum is attained by a union of Π -clusters, and by Lemma 1(iii) at $\Pi_{\perp}$ the unrestricted minimum is min over Eq. (1)’s exploration set — exactly the stated condition. $\square$

Proof of Prop. 4. We work in the SCM with exogenous variables $\mathbf{U}$ represented so that all elements of $\mathbf{U}$ are mutually independent and every latent common cause is a single shared exogenous variable (a bidirected edge in $\mathcal{G}^*$); this is without loss of generality for semi-Markovian models.

Fix a cluster C , a strict subset $\mathbf{X}_A \subsetneq C$, a value $\mathbf{a}$, and write $\mathbf{B} = C \setminus \mathbf{X}_A$. Let $\mathbf{B}^{\text{nat}}(\mathbf{u})$ be the value $\mathbf{B}$ takes under $\text{do}(\mathbf{X}_A = \mathbf{a})$ at exogenous realization $\mathbf{u}$. By the consistency of interventions, setting $\mathbf{B}$ to the value it would have taken changes nothing realization by realization, so $Y_{\mathbf{u}}(\text{do}(\mathbf{a})) = Y_{\mathbf{u}}(\text{do}(\mathbf{a}, \mathbf{B} = \mathbf{B}^{\text{nat}}(\mathbf{u})))$.

The independence step, made explicit. Under $\text{do}(\mathbf{X}_A = \mathbf{a})$, $\mathbf{B}^{\text{nat}}$ is a measurable function of $\mathbf{U}_{\mathbf{B}}$, the exogenous variables entering the mechanisms of $\mathbf{B}$ and of its non-intervened ancestors. At fixed $(\mathbf{a}, \mathbf{b})$, $Y_{\mathbf{u}}(\text{do}(\mathbf{a}, \mathbf{b}))$ is a measurable function of $\mathbf{U}_{\mathbf{Y}}$, the exogenous variables entering the mechanisms of Y and of its ancestors outside C (all of C is intervened in this world, so no path into Y passes through C). Suppose some $U^* \in \mathbf{U}_{\mathbf{B}} \cap \mathbf{U}_{\mathbf{Y}}$. Then U^* has a directed path into some member of $\mathbf{B}$ — take b^* to be the *first* cluster member on that path, so the segment $U^* \rightarrow \dots \rightarrow b^*$ meets C only at its endpoint — and a directed path into Y avoiding C entirely. Concatenating them tail-to-tail yields $b^* \leftarrow \dots \leftarrow U^* \rightarrow \dots \rightarrow Y$: a backdoor path from the cluster member b^* to Y whose interior avoids C , i.e. a backdoor path escaping the cluster, which premise (ii) excludes. Hence $\mathbf{U}_{\mathbf{B}}$ and $\mathbf{U}_{\mathbf{Y}}$ are disjoint sets of mutually independent exogenous variables, so $\mathbf{B}^{\text{nat}}$ is independent of the map $\mathbf{u} \mapsto Y_{\mathbf{u}}(\text{do}(\mathbf{a}, \mathbf{b}))$ for every fixed $(\mathbf{a}, \mathbf{b})$.

Taking expectations therefore factorizes into a mixture, $\mathbb{E}[Y \mid \text{do}(\mathbf{a})] = \int \mathbb{E}[Y \mid \text{do}(\mathbf{a}, \mathbf{b})] dP(\mathbf{B}^{\text{nat}} = \mathbf{b} \mid \text{do}(\mathbf{a}))$, and a mixture is bounded below by its best component. Premise (i) makes that component reachable: the whole-cluster arm may set $\mathbf{b}$ anywhere in the natural support, so $\mu^*(C) \leq \mu^*(\mathbf{X}_A)$. Applying the argument per cluster covers arms spanning several clusters, hence $V(\Pi) = V(\Pi_{\perp})$, and the minimizing whole-cluster values witness a union-of-clusters optimizer.

The two examples below are not decoration: they show the premises are *tight* — dropping either one admits an SCM with strictly positive price, so no weaker hypothesis of this form can give the conclusion.

Necessity of (i), realized by MEDIATEDCHAIN (§5.3). The policy box restricts hard interventions on X_2 to $[-2, 2]$ while the mechanism $X_2 = 2X_1$ drives X_2 ’s natural response to 4, the bowl center of $Y = (X_2 - 4)^2$, when $X_1 = 2$ is set. The mediated path reaches values the box forbids, so $\text{do}(X_1 = 2)$ strictly beats every in-box whole-cluster intervention: the price is exactly $V(\Pi) - y^* = (2 - 4)^2 - \sigma_2^2 = 3.96$ (App. B). (The mixture argument fails at the reachability step: the best component $\mathbf{b}$ lies outside the interventional domain.)

Necessity of (ii). Let $C = \{A, B\}$ with latent U , mechanisms $B = U$ and $Y = -BU + (A - 1)^2$, U standard normal. Under $\text{do}(A = 1)$ with B free, $\mathbb{E}[Y] = -\mathbb{E}[U^2] = -1$. Under any whole-cluster $\text{do}(A = a, B = b)$, $\mathbb{E}[Y] = -b\mathbb{E}[U] + (a - 1)^2 \geq 0$. Intervening severs the exploitable correlation between B and Y : the backdoor path

$B \leftrightarrow Y$ escapes the cluster, $U \in \mathbf{U}_B \cap \mathbf{U}_Y$, the independence step fails, and the price is 1 despite unconstrained domains. $\square$

Proof of Remark 6. Split the run at the trigger. Pre-trigger, every graph-dependent component factors through $(\Pi, \mathcal{G}^\Pi)$, and by Remark 3 the trigger time and split decision are functions of the trajectory and the declared hierarchy alone, so two hypotheses with the same $\mathcal{G}^\Pi$ produce the same phase-1 run *and* the same refined partition Π' . Post-trigger, the components factor through $(\Pi', \mathcal{G}^{\Pi'})$, so Theorem 1 applies verbatim at Π' : hypotheses agreeing on $\mathcal{G}^{\Pi'}$ are indistinguishable. An edit confined to a never-split cluster is invisible in both $\mathcal{G}^\Pi$ and $\mathcal{G}^{\Pi'}$ (the cluster is intact in both partitions), hence protected across the whole run; an edit inside the split cluster C is invisible in $\mathcal{G}^\Pi$ but generally visible in $\mathcal{G}^{\Pi'}$ (its endpoints now lie in distinct Π' -clusters), which is exactly the exposure purchased. No other edit changes visibility: quotient edges between untouched clusters are the same in $\mathcal{G}^\Pi$ and $\mathcal{G}^{\Pi'}$. $\square$

B MinimalBench: specification and oracles

The three SCMs share one parameterization — Gaussian roots, manipulable domains $[-3, 3]$ (except the MEDIATEDCHAIN policy box), unit intervention costs, target noise $\sigma_Y = 0.1$, no clipping — so every arm value is closed-form. The structural module that defines them (`ccbo/minibench.py`) is the same one the methods, the perturbation registry, the structural tests, and Table 2’s emitter consume, so specification and experiment cannot drift. Observational datasets (5,000 rows, seed 0; every run uses the first $n_{\text{obs}}=100$) are produced by released generators whose built-in oracle gates assert the Monte-Carlo arm values against the closed forms below (tolerance 0.05) at build time.

PARALLEL PARENT (Experiment A). Latent U ; manipulable X_1, X_2 ; target Y :

$$U = 0.4 \epsilon_0, \quad X_i = U + 0.3 \epsilon_i, \quad Y = \lambda(X_1 - 2)^2 + (X_2 + 2)^2 + \sigma_Y \epsilon_Y,$$

with $\lambda = 1$; the latent projection carries $X_1 \leftrightarrow X_2$ (corr = 0.64), and $\text{Var} X_i = 0.25$. Coarse partition $C_1 = \{X_1, X_2\}$. Closed forms: $\mathbb{E}[Y \mid \text{do}(x_1, x_2)] = \lambda(x_1 - 2)^2 + (x_2 + 2)^2$, so $y^* = 0$ at the joint arm; $\mathbb{E}[Y \mid \text{do}(x_1)] = \lambda(x_1 - 2)^2 + 4.25$ (the free co-parent’s bowl plus its variance) and $\mathbb{E}[Y \mid \text{do}(x_2)] = 4.25\lambda + (x_2 + 2)^2$. Under A1 (delete $X_1 \rightarrow Y$) the fine MIS is $\{X_2\}$ alone — $X_1 \notin \text{An}(Y)$ in every mutilation — so the best reachable fine value is $G(\lambda) = \lambda(4 + 0.25) = 4.25\lambda$, the analytic gap of §5.1. Conditions: A1 = del $X_1 \rightarrow Y$ (quotient-redundant); A2 = add $X_1 \rightarrow X_2$ (intra-cluster bow with the latent $X_1 \leftrightarrow X_2$: $\text{do}(X_1)$ loses identification, membership intact); A3 = assume $X_1 \leftrightarrow Y$ (quotient-visible: the C-DAG gains $C_1 \leftrightarrow Y$ and the cluster arm drops to the uninformative tier; QCBO keeps running).

FRONT DOOR (Experiment B). Latent U ; manipulable X_1, M ; target Y :

$$U = 0.4 \epsilon_0, \quad X_1 = U + 0.3 \epsilon_1, \quad M = 2X_1 + 0.2 \epsilon_M, \\ Y = 2(1 - e^{-(M-1)^2/(2 \cdot 0.3^2)}) + U + \sigma_Y \epsilon_Y.$$

The latent projection carries $X_1 \leftrightarrow Y$. The target is a narrow well (width 0.3 over the 6-wide domain): prior corruption is a global-search phenomenon, and on a smooth quadratic the same corruption measurably costs nothing (§5.2). Fine MIS = $\{X_1\}, \{M\}$ (chain); $\text{do}(M)$ is identified by backdoor through X_1 , $\text{do}(X_1)$ by the front-door functional through M . Gaussian smoothing gives the closed forms: $\mathbb{E}[Y \mid \text{do}(m)] = 2(1 - e^{-(m-1)^2/0.18})$, so $y^* = 0$ at $\text{do}(M=1)$; $\mathbb{E}[Y \mid \text{do}(x_1)] = 2(1 - \frac{0.3}{\sqrt{0.3^2 + 0.2^2}} e^{-(2x_1-1)^2/(2(0.3^2+0.2^2))})$, best ≈ 0.336 at $x_1 = 1/2$; observational mean $\mathbb{E}[Y] \approx 1.637$. The coarse partition $\{X_1, M\}$ quotients to the bow $C_1 \rightarrow Y$, $C_1 \leftrightarrow Y$: the cluster arm runs on the uninformative tier in *every* condition, and setting M directly makes the partition lossless ($V(\Pi) = y^* = 0$). Condition B1 assumes a spurious intra-cluster confounder $X_1 \leftrightarrow M$: both fine arms lose identification (the backdoor set $\{X_1\}$ for $\text{do}(M)$ is invalidated by the induced collider path, the front-door premise for $\text{do}(X_1)$ by the confounded $X_1 \rightarrow M$ edge), while the quotient — which drops intra-cluster bidirected edges — is unchanged.

MEDIATED CHAIN (Experiment C). No latents; manipulable X_1, X_2 ; target Y :

$$X_1 = 0.5 \epsilon_1, \quad X_2 = 2X_1 + 0.2 \epsilon_2, \quad Y = (X_2 - 4)^2 + \sigma_Y \epsilon_Y,$$

with $D(X_1) = [-3, 3]$ and the *policy box* $D(X_2) = [-2, 2]$. Fine MIS = $\{X_1\}, \{X_2\}$: the joint is not minimal on a chain (mutilating X_2 severs X_1 from $\text{An}(Y)$; Prop. 2), so the joint intervention exists only as the quotient’s cluster arm, which must clamp X_2 into the box. Closed forms: $\mathbb{E}[Y \mid \text{do}(x_1)] = (2x_1 - 4)^2 + 0.04$, so $y^* = 0.04$ at $\text{do}(X_1=2)$ — driving X_2 ’s natural response to 4, outside the box; $\mathbb{E}[Y \mid \text{do}(x_2)] = (x_2 - 4)^2$, best over the box = 4; hence $V(\Pi) = 4$ and the price $V(\Pi) - y^* = 3.96$ (Prop. 3; premise (i) of Prop. 4 fails by construction). The declared hierarchy splits $C_1 = \{X_1, X_2\}$ into singletons; one split reaches the identity partition, whose MIS contains the optimal $\text{do}(X_1)$ arm.

Structural tests and oracle gates. Dataset-free structural tests assert, per condition, the exact fine MIS, the C-DAG signature (in)equalities, and the prior-tier flips claimed above, and end-to-end regressions assert QCBO’s trajectory invariance (arm choices and intervention values exactly; incumbents to 10^{-8}) under every protected condition and its divergence under A3. The data generators refuse to write datasets whose Monte-Carlo arm values, correlations, or policy-box geometry contradict the closed forms.

C Fixed-objective protocol: per-method details

The invariant is that every method optimizes the *same* objective, the true SCM’s noise-free `intervention_function`, and only the *structure each method reasons with* is perturbed. We realize this per method as follows.

QCBO (finest and coarse). The objective is read from the original graph’s SEM, while the structure is supplied separately through the assumed graph name (`CoarsenedGraph(..., assumed_graph_name=variant)`). Swapping the assumed graph changes the quotient, the exploration set, the identification formulas, and the prior, but never the SEM that generates outcomes. *Prior pipeline.* An identifiable arm’s prior mean instantiates the identified formula as a pipeline of GP-based plug-ins (backdoor $\rightarrow$ front-door $\rightarrow$ g -computation) on the bidirected-edge expansion of $\mathcal{G}^\Pi$. A non-identifiable arm — and any identifiable query the sound-but-incomplete pipeline cannot instantiate — shares the common uninformative prior, the observational mean of Y . Because clusters are manipulable-only and interventions assign full clusters, the cluster-level formulas expand exactly into fine-grained regressions on observational data.

CBO (full-DAG identification). CBO is QCBO’s backend at the identity partition (Prop. 5): it runs the same complete identification algorithm, only on the full DAG rather than a quotient. The injection is the same assumed-graph swap as for QCBO (the perturbed structure reaches the exploration set and the identification formulas that build the prior), while outcomes are always drawn from the true SEM. Its trajectory is identical to QCBO’s at the finest partition on every graph.

Exploration-set disclosure. Both methods use the MIS of the assumed graph (Eq. 2), computed by the released code and verified against hand-derived Lee–Bareinboim ground truth per benchmark (for the minimal suite, the per-condition sets are also asserted by the structural tests of App. B). The CBO authors’ shipped code deviates from that definition in two places, which we correct for both methods in the family suite: on ToyGraph it lists the non-minimal arm $\{X, Z\}$ (mutilating Z cuts $X \rightarrow Z$, so $\text{do}(X, Z) \equiv \text{do}(Z)$ and the joint arm is redundant), and on CoralGraph it truncates the list at subsets of size three (an ad-hoc cap; the definition yields all 31 non-empty subsets, as every manipulable variable keeps a directed path to Y through never-intervened non-manipulables under any mutilation).

HQCBO. Phase 1 is the QCBO configuration run by the same code path with the same seed (the trigger is a function of the coarse trajectory alone), so its pre-trigger prefix is byte-identical to the QCBO run — asserted per seed. The refined phase re-enters the same backend with the split partition, the unioned exploration set, and the carried per-arm data (§3.2); the vendored loop forces an observation step at the start of a phase, a small bias *against* refinement that we keep.

Objective-identity verification. For all methods the decoupling holds by construction: outcomes are always evaluated on the true graph’s SEM, and the perturbation enters only through the separately-supplied assumed-graph variant (including assumed latents, as in A3 and B1). The released data generators additionally refuse to build any dataset whose Monte-Carlo arm values contradict the closed-form oracles (App. B): feeding a method a wrong graph changes only its causal reasoning, never the problem it is solving.



Figure 7: **ParallelParent (MinimalBench, Exp. A)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. The intra-cluster $X_1 \leftrightarrow X_2$ disappears in the quotient; deleting $X_1 \rightarrow Y$ leaves $C_1 \rightarrow Y$ alive through $X_2 \rightarrow Y$ (quotient-redundant).



Figure 8: **FrontDoor (MinimalBench, Exp. B)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. The quotient is a bow ($C_1 \rightarrow Y$, $C_1 \leftrightarrow Y$): the cluster arm runs on the uninformative prior tier in every condition, so the assumed intra-cluster $X_1 \leftrightarrow M$ cannot change it.

D Dataset graphs and their quotients

For every dataset in the paper, the figures below show the fine DAG with its coarse partition and the quotient C-DAG the method actually consumes — regenerated from the released structure registry by `scripts/emit_dataset_tikz.py`, so the pictures cannot drift from the code. On **PARALLELParent**, note how the intra-cluster latent $X_1 \leftrightarrow X_2$ disappears in the quotient while $C_1 \rightarrow Y$ survives any single parent-edge deletion — the redundancy behind Experiment A; the **FRONTDOOR** and **ToyGraph** pairs both quotient to the bow of Fig. 3 — the exposure trade of Prop. 7.

E Implementation and reproducibility

All methods share one surrogate/acquisition backend (the vendored CBO loop), so comparisons differ only in the assumed structure. **Surrogate**. Each exploration-set arm carries a `GPY GPRegression` model with an RBF kernel (signal variance and lengthscale initialized to 1, ARD off) and observation noise fixed near zero (10^{-10} , matching the deterministic mechanisms). The causal prior supplies the do-calculus estimate as the GP *mean function*, so Δ -metrics isolate the effect of a wrong prior/exploration set from surrogate noise. **Acquisition**. Causal expected improvement [Garnett, 2023], maximized over the box-constrained intervention domain by gradient-based L-BFGS. **Protocol (minimal suite)**. Unit per-variable intervention costs, 3 initial interventional points per arm, $T=50$ trials (60 on **MEDIATEDCHAIN**), 30 seeds, $n_{\text{obs}}=100$ from one shared dataset per SCM; the family suite follows each base paper’s own protocol (§5.4). Every run additionally logs the selected arm and intervention values per trial, so invariance is checked on *decisions*, not only on incumbent values. **Invariance tolerance**. Arm-choice and intervention-value sequences are compared exactly; incumbent trajectories at 10^{-8} (GP-solver floating point). On every protected condition of Table 2 the sequences are identical and $\sup |\Delta| = 0$; the quotient-visible control moves trajectories on every seed. **Artifacts**. The MinimalBench structural module (`ccbo/minibench.py`), its data generators with built-in oracle gates, the dataset-free structural tests and end-to-end invariance regressions, the suite runner (`scripts/run_minimal_suite.py`), the native two-phase HQCBO



Figure 9: **MediatedChain (MinimalBench, Exp. C)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. The policy box restricts $D(X_2)$ to $[-2, 2]$; the fine optimum $\text{do}(X_1=2)$ drives X_2 to 4, outside the box.



Figure 10: **ToyGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. The quotient is a bow ($C_1 \rightarrow Y$, $C_1 \leftrightarrow Y$): $\text{do}(C_1)$ is not identifiable from the C-DAG, so the arm runs on the uninformative prior tier.

(ccbo/minimal_suite.py), and every table/figure emitter are released, so all reported numbers regenerate from the released SCMs and seeds.

F Full results

Table 5 positions QCBO in the method landscape (§6). Tables 4 and 6 give the per-environment detail behind §5.4’s model-based rows: finals and wall-clock on MCBO’s own tasks, and the within-family invariance counts under the intra-cluster perturbation of the model’s graph view. Per-seed CSVs for every minimal-suite unit (trial, incumbent, cumulative cost, selected arm, and intervention values) regenerate from the released runners (App. E).

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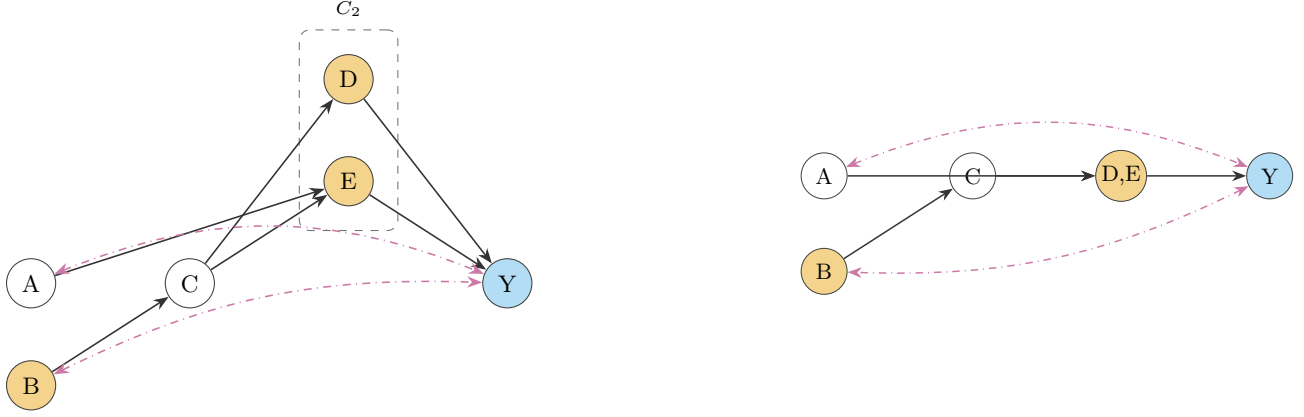


Figure 11: **CompleteGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

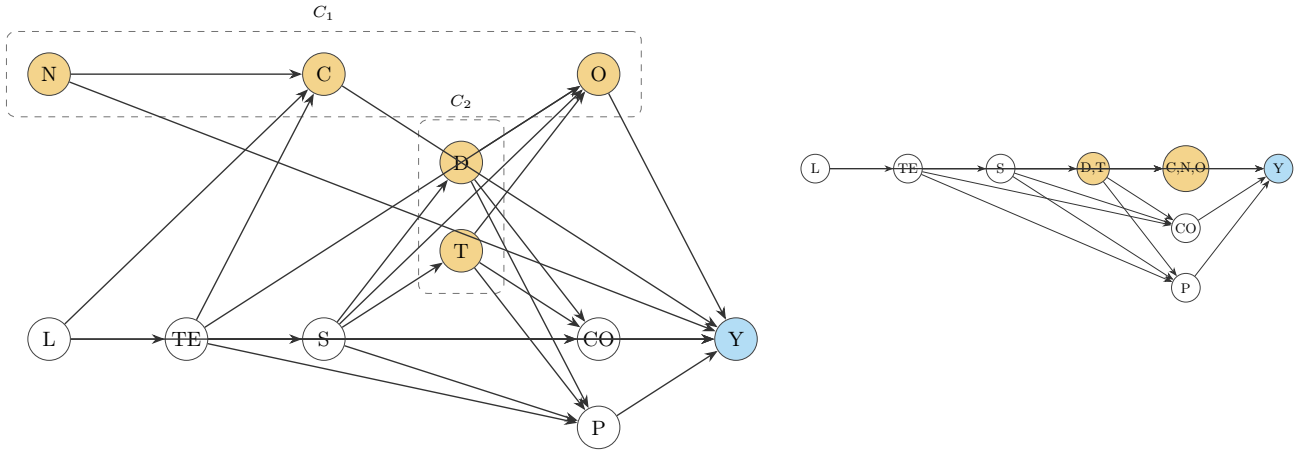


Figure 12: **SimplifiedCoralGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

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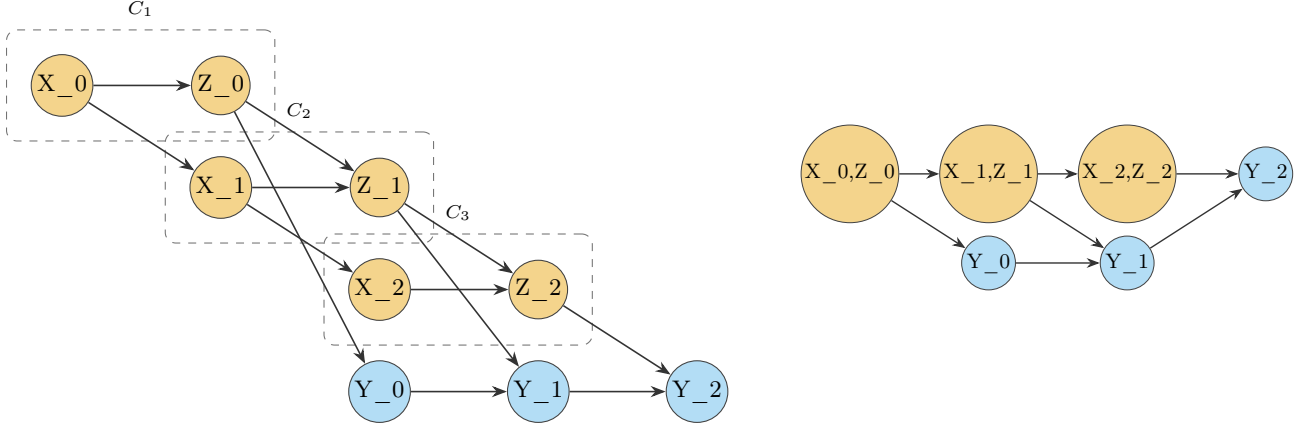


Figure 13: **DCBO stat / nonstat (QDCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. Slices $t = 0, 1, 2$; the nonstat setup shares this topology with a change point in the SEM.

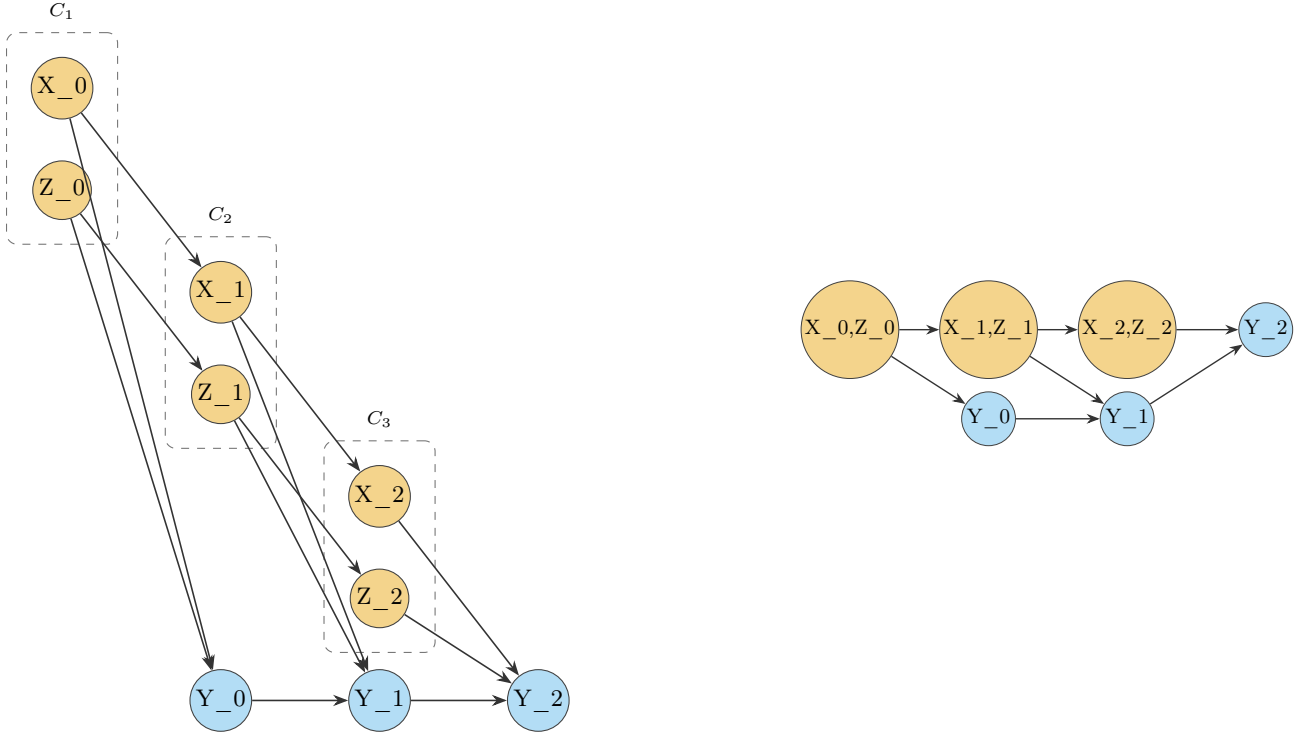


Figure 14: **DCBO ind (QDCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

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Figure 15: **ToyGraph-M (QMCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

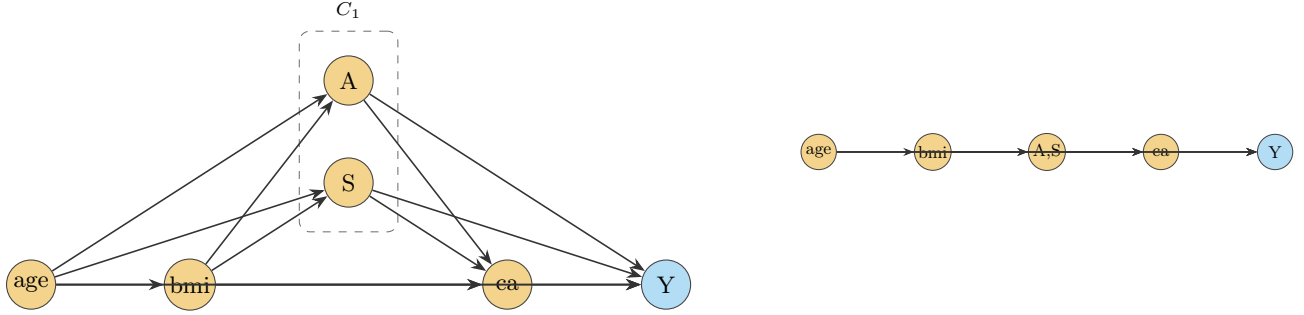


Figure 16: **PSAGraph (QMCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

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Algorithm 1 QCBO: Quotient Causal Bayesian Optimization

Require: C-DAG $\mathcal{G}^\Pi$ over the clusters of Π (or a fine hypothesis $\mathcal{G}$ and Π); target Y ; number of trials T ; observational data $\mathcal{D}^O$; optional hierarchy H (§3.2).

- 1: If given $\mathcal{G}$: $\mathcal{G}^* \leftarrow$ latent projection of $\mathcal{G}$; $\mathcal{G}^\Pi \leftarrow$ quotient of $\mathcal{G}^*$ under Π . Check acyclicity of $\mathcal{G}^\Pi$ (Assumption 1).
- 2: Build $\mathbf{ES}^\Pi \leftarrow \text{MIS}(\mathcal{G}^\Pi)$ via Eq. (3).
- 3: For each $\mathbf{X}_s \in \mathbf{ES}^\Pi$: run complete ID on $P(Y \mid \text{do}(\mathbf{X}_s))$ and fit the prior mean m_s via the two-tier rule, Eq. (4), on $\mathcal{D}^O$.
- 4: Collect initial interventional data $\mathcal{D}_0^I$.
- 5: **for** $t = 1, \dots, T$ **do**
- 6: Update GP posteriors $(m_{s,t}, k_{s,t})$ for all $\mathbf{X}_s \in \mathbf{ES}^\Pi$.
- 7: $(\mathbf{X}_{s_t}, \mathbf{x}_t) \leftarrow \arg \max_{s, \mathbf{x}} \text{EI}_t^s(\mathbf{x}) / \text{Co}(\mathbf{X}_s, \mathbf{x})$.
- 8: Intervene, observe y_t ; update $\mathcal{D}_t^I$.
- 9: **if** H declared **and** incumbent plateaued: split the incumbent’s clusters via H , re-check validity, rebuild $\mathbf{ES}^\Pi \cup$ past arms, carry $\mathcal{D}_t^I$ by arm, init new arms (§3.2).
- 10: **end for**
- 11: **return** $(\mathbf{X}_{s_{t^*}}, \mathbf{x}_{t^*})$ where $t^* = \arg \min_t y_t$.

 Table 4: QMCBO vs MCBO (its own harness and protocol; 5 seeds, $T=100$). Final Y (all tasks maximized) and wall-clock per unit.

Env	Method	Final Y	s/unit
ToyGraph ($y^*=2.172$)	MCBO	1.143 ± 0.271	1410
	QMCBO	2.150 ± 0.013	1570
PSAGraph ($y^* = -4.13$)	MCBO	-5.153 ± 0.000	6830
	QMCBO	-5.153 ± 0.000	2266

Table 5: Positioning of QCBO against the nearest methods.

Method	What is learned	Structural prior	Search space
CBO [Aglietti et al., 2020]	nothing (oracle DAG)	full DAG	MIS
GACBO [Mukherjee et al., 2024]	posterior over DAGs	DAG family	union of MIS
Durand et al. [Durand et al., 2025]	$\text{Pa}(Y) \subseteq \mathbf{X}$	Y leaf	$\mathbf{X}$
Park et al. [Park et al.]	$\text{PAG} \rightarrow \text{MAG}$	MEC	partial $\mathbf{X}$
SCB [Lee and Bareinboim, 2018]	nothing (oracle DAG)	full DAG	POMIS
QCBO (ours)	nothing	partition + C-DAG	identifiable cluster unions

 Table 6: Invariance under an intra-cluster misspecification of the model’s graph view (same seeds, deterministic runs): trajectories identical to the correct-graph run, and mean $|\Delta_{\text{final}}|$. QMCBO cannot move (Theorem 1). MCBO does.

Env	Method	identical	$ \Delta_{\text{final}} $
ToyGraph	MCBO	2/5	0.299
	QMCBO	5/5	0.000
PSAGraph	MCBO	0/5	0.000
	QMCBO	5/5	0.000